

Vanessa Tan

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in vtan25

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Education

- 2022 – Present  **Ph.D. in Culture Technology, KAIST**
Topic: *Cross-Modal Music–Dance Generation and Cross-Domain Motion Transfer*
Advisor: Prof. Junyong Noh
- 2021 – 2022  **Korean Language Program, Duksung Women's University**
Global Korea Scholarship (GKS) Awardee; TOPIK Level 3
- 2016 – 2019  **M.Sc. in Electrical Engineering, University of the Philippines Diliman**
Topic: *Multi-Task Learning for Detection, Recovery, and Separation of Polyphonic Music*
Advisor: Prof. Franz de Leon
- 2010 – 2015  **B.S. in Electronics and Communications Engineering, University of the Philippines Diliman**
Topic: *Melody Extraction for Philippine Indigenous Music*
Advisor: Prof. Franz de Leon

Work Experience

- August 2022 – Present  **Research Assistant, Visual Media Lab (KAIST)**
Project 1: Controllable Generative UV Texture Synthesis for Digital Avatars
Project 2: Cross-Morphology Motion Retargeting and Transfer
Advisor: Prof. Junyong Noh
- January 2025 – June 2025  **Research Intern, Sony CSL Japan**
Dance-Music Correspondence Learning
Advisor: Taketo Akama and Natalia Polouliakh
- August 2019 – August 2021  **University Researcher, Philippine Space Agency**
Development of Mission Attitude and Telemetry Analysis Software with Machine Learning–Based Cloud Detection and Satellite Attitude Control
Advisor: John Leur Labrador and Prof. Marc Talampas
- August 2018 – July 2019  **Research Assistant, Digital Signal Processing Lab (UP Diliman)**
Digital Archiving and Analysis of Philippine Indigenous Instruments
Advisor: Prof. Franz de Leon
- February 2018 – July 2018  **Freelance Unity Developer, Imagine Realities Inc.**
Led End-to-End Software Delivery and QA for Client Event Application
- April 2016 – April 2018  **Research Associate, Ubiquitous Computing Lab (UP Diliman)**
Project 1: VR Scene Creation Platform with Automated Skeletal Rigging and 3D Motion Dataset Development
Project 2: Immersive Scene Design for VR Fantasy Environments
Advisor: Prof. Rowel Atienza
- September 2015 – March 2016  **Associate Software Engineer, Accenture Inc.**
Customer-Focused Onboarding and Technical Solutions for Office 365 in Collaboration with Offshore Engineering Team

Research Publications

- 1 S. Jung, Y. Seol, K. Seo, H. Na, S. Kim, **V. Tan**, and J. Noh, “Speed-aware audio-driven speech animation using adaptive windows,” *ACM Trans. Graph.*, vol. 44, no. 1, Oct. 2024, ISSN: 0730-0301.

- 2 J.-A. R. Sarmiento, **V. H. Tan**, M. C. R. Talampas, and P. C. Naval Jr, "Sample efficient deep reinforcement learning for diwata microsatellite reaction wheel attitude control," *Aerospace Systems*, vol. 6, no. 1, pp. 61–69, 2023.
- 3 **V. Tan**, J. Nam, J. Nam, and J. Noh, "Motion to dance music generation using latent diffusion model," in *SIGGRAPH Asia 2023 Technical Communications*, ser. SA '23, Sydney, NSW, Australia: Association for Computing Machinery, 2023, ISBN: 9798400703140.*
- 4 **V. Tan**, J. A. Banatao, J. L. Labrador, L. C. Mabaquiao, F. F. Fortes, and M. C. Talampas, "Mata-cloud: A cloud detection and dynamic attitude correction evaluation software," in *Small Satellite Conference*, 2021.
- 5 **V. Tan**, J. L. Labrador, and M. C. Talampas, "Mata-rl: Continuous reaction wheel attitude control using the mata simulation software and reinforcement learning," in *Small Satellite Conference*, 2021.
- 6 **V. Tan**, J. L. Labrador, and M. Caesar Talampas, "Mata: Mission, attitude, and telemetry analysis software for micro-satellites," in *2020 IEEE REGION 10 CONFERENCE (TENCON)*, 2020, pp. 614–619.*
- 7 **V. Tan** and F. d. Leon, "Multi-task learning for detection, recovery, and separation of polyphonic music," in *2020 IEEE REGION 10 CONFERENCE (TENCON)*, 2020, pp. 1112–1117.
- 8 **V. H. Tan** and F. de Leon, "Time-frequency representations for single-channel music source separation," in *2019 International Symposium on Multimedia and Communication Technology (ISMATC)*, 2019, pp. 1–6.
- 9 R. Blonna, M. S. Tan, **V. Tan**, A. P. Mora, and R. Atienza, "Vrex: A framework for immersive virtual reality experiences," in *2018 IEEE Region Ten Symposium (Tensymp)*, 2018, pp. 118–123.
- 10 J. M. Mendoza, **V. Tan**, V. Fuentes, G. Perez, and N. M. Tiglaio, "Audio event detection using wireless sensor networks based on deep learning," in *International Wireless Internet Conference*, Springer International Publishing Cham, 2018, pp. 105–115.*
- 11 **V. H. Tan**, R. O. Atienza, M. I. Saldares, J. Casimiro, and M. S. Viola, "Aneeme: Synthesizing and sharing animation building blocks for rapid creation of 3d virtual scenes," in *SIGGRAPH Asia 2017 VR Showcase*, ser. SA '17, Bangkok, Thailand: Association for Computing Machinery, 2017, ISBN: 9781450354080.
- 12 J. Disuanco, **V. Tan**, and F. de Leon, "Study of automatic melody extraction methods for philippine indigenous music," in *2015 IEEE International Conference on Control System, Computing and Engineering (ICCSCE)*, 2015, pp. 464–469.*

* Co-first authors (equal contribution).

Skills

Programming	Python, Pytorch, Keras, Tensorflow, C#, MATLAB, LaTeX, Git
Simulation Software	Unity3D, Unreal Engine, Motionbuilder, Blender
Certifications	Licensed Philippine Electronics Engineer, Creative Applications of Deep Learning with Tensorflow (Kadenze)
Languages	Filipino (Native), English (Fluent), Korean (Intermediate), Japanese (Basic)
Others	Machine Learning, Deep Learning, Project Management, Virtual Reality, Game Development

Miscellaneous Experience

Awards and Scholarships

- 2021 - 2025  **Global Korea Scholarship**, National Institute for International Education.
- 2018 - 2019  **DOST-SEI Scholar**, Engineering Research and Development for Technology.
- 2018  **Finalist (Included in the Top 12)**, Ideospace Startup Competition.
-  **Second Runner-Up**, AWS Hackdays Manila (Machine Learning Edition).

Miscellaneous Experience (continued)

- 2015  **Best Paper**, 8th AUN/SEED-NET Conference.
- 2014  **University Scholar (Dean's Lister)**, University of the Philippines Diliman

Activities

- 2023 - Present  **Student Ambassador**. KAIST Graduate Admissions Team.
- 2026  **Reviewer**. ACM CHI Posters.
- 2023  **Reviewer**. IEEE Region 10 Conference (TENCON).
- 2021  **Contributor**. Space and Artificial Intelligence Conference.
-  **Speaker**. COE197D: Data Driven Control.
- 2020  **Reviewer**. IEEE Region 10 Conference (TENCON).
-  **Student Volunteer**. SIGGRAPH Asia.
- 2019  **Speaker**. UX of XR - Designing for Immersive Technologies.
- 2018  **Speaker**. Computer Research and Engineering Symposium (CoRES).
- 2017  **Speaker**. Animahenasyon: 11th Philippine Animation Festival.

References

Available on Request